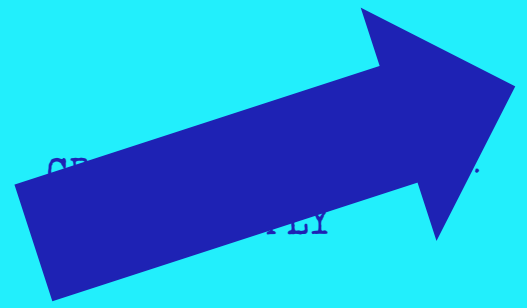
AGRICULTURAL MARKET PRICE ANALYSIS

MANDI PRICE PULSE

When should a farmer sell?

Using SQL and Power BI



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THE DECISION FRAME

THREE VERIFIED SIGNALS FRAME THE SELLING DECISION

Use history to compare the window, not to promise the outcome.



Coverage

17,496 records spanning 2021–2025 across onion, tomato, and wheat.



Volatility

Tomato is highest at 45.8% CV; wheat is lowest at 9.3%.



Seasonality

Onion peaks in November, tomato in July, wheat in February.

> Decision supported: compare crop, month, market, and supply conditions. Historical patterns are not a guaranteed forecast.

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THE PRACTICAL QUESTION

“WHEN SHOULD I SELL?”

A DECISION-SUPPORT PROBLEM

Prices shift by crop, month, market, and supply conditions. The practical question is not “what is the highest price ever?” but “what is a defensible selling window for this crop, in this market, given storage and transport constraints?”

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DATA FOUNDATION

THE DATASET SUPPORTS CROP, MARKET, PRICE, AND SUPPLY ANALYSIS

Dimension	Available evidence	Why it matters
Fields	Arrival_Date, Commodity, State, District, Market, Variety, Grade, Min_Price, Max_Price, Modal_Price, Arrivals_Qtl	Join price movement to crop, place, quality, and supply.
Scope	17,496 records · 3 commodities · 9 states · 11 markets	January 1, 2021 to December 31, 2025.
Validation	Currency and unit labels are not specified in the supplied CSV.	Confirm before external publication.

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ANALYTICAL PIPELINE

SQL TURNS RAW MANDI RECORDS INTO DECISION-READY EVIDENCE

01

Raw agricultural records

02

SQL analysis and validation

03

Monthly, volatility, market, and supply outputs

04

Power BI dashboard and filters

05

Conditional selling recommendations

Tools used: SQL + Power BI.

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BUSINESS QUESTIONS

THE ANALYSIS ANSWERS TEN PRACTICAL BUSINESS QUESTIONS

Timing

Monthly average prices, best and worst months, and best consecutive three-month windows.

Risk

Coefficient of variation, price spread, floors, peaks, and peak-to-floor ratios.

Market and supply

Spikes and crashes, year-on-year change, market rankings and stability, plus arrivals-versus-price association.

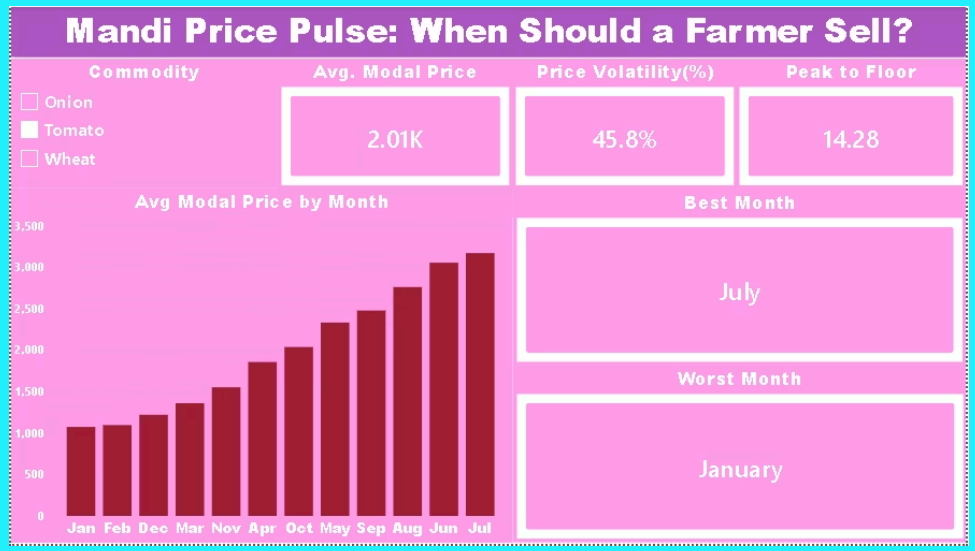
> Arrivals-versus-price patterns are interpreted as association rather than causation.

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DECISION INTERFACE

THE DASHBOARD MAKES THE SELLING WINDOW VISIBLE

Filters turn the analysis into a crop-and-market conversation rather than a single headline number.



> Displayed example: Avg Modal Price 2.01K · Price Volatility 45.8% · Peak to Floor 14.28

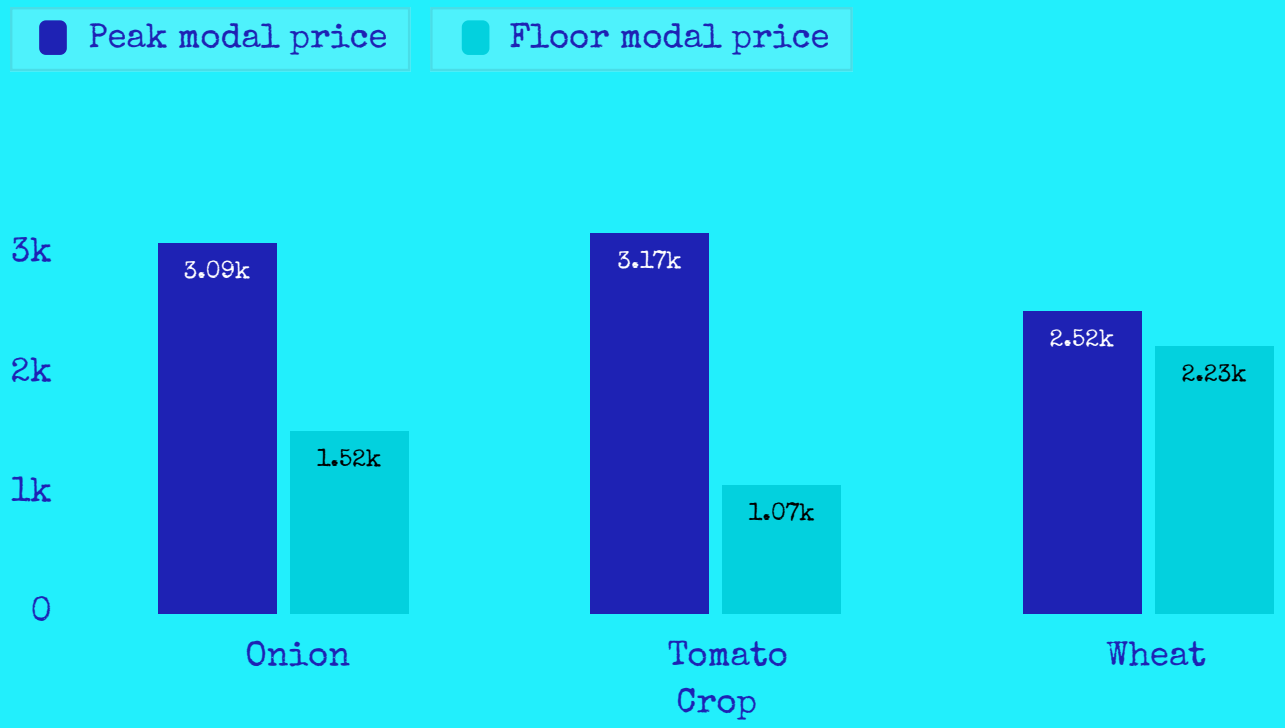
Best Month July · Worst Month January

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SEASONALITY

SEASONALITY DIFFERS BY CROP, SO ONE CALENDAR CANNOT FIT ALL

Verified extrema in monthly average modal prices, 2021–2025



> Onion peaks in November and bottoms in April.

Tomato peaks in July and bottoms in January.

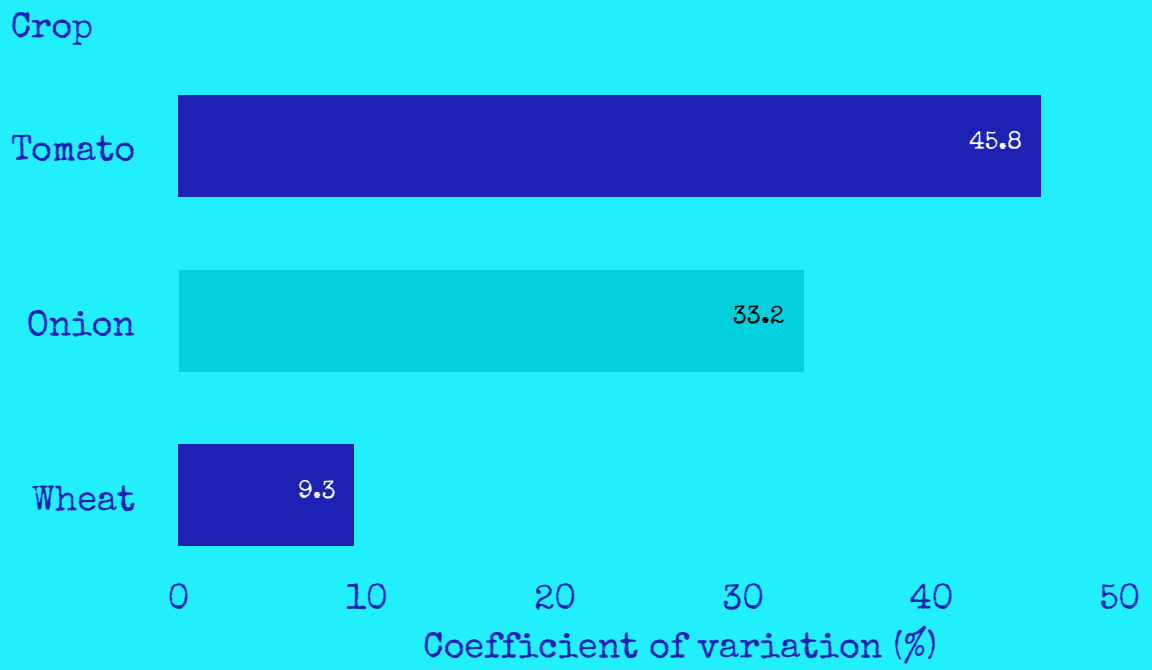
Wheat peaks in February and bottoms in April.

Best three-month windows require a defined rolling-window output before publication.

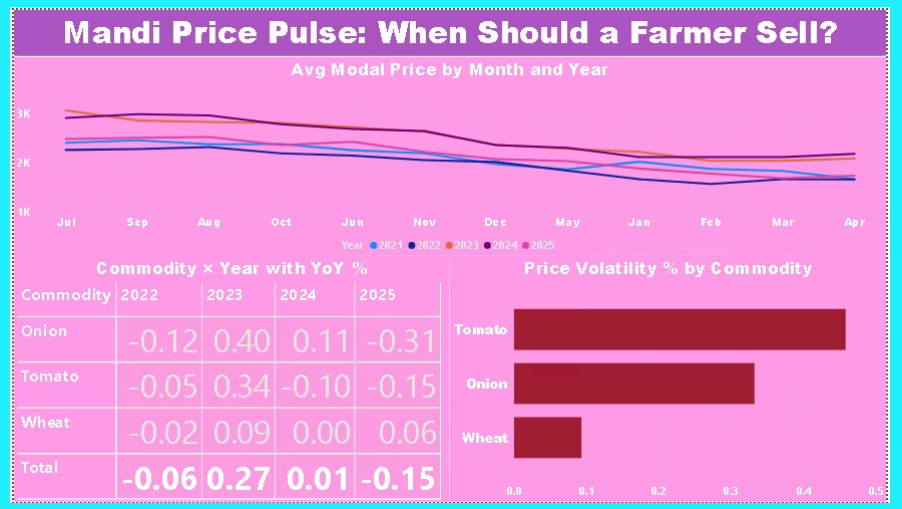
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RISK OF WAITING

VOLATILITY CHANGES THE RISK OF WAITING



Higher CV means a less stable historical price pattern.



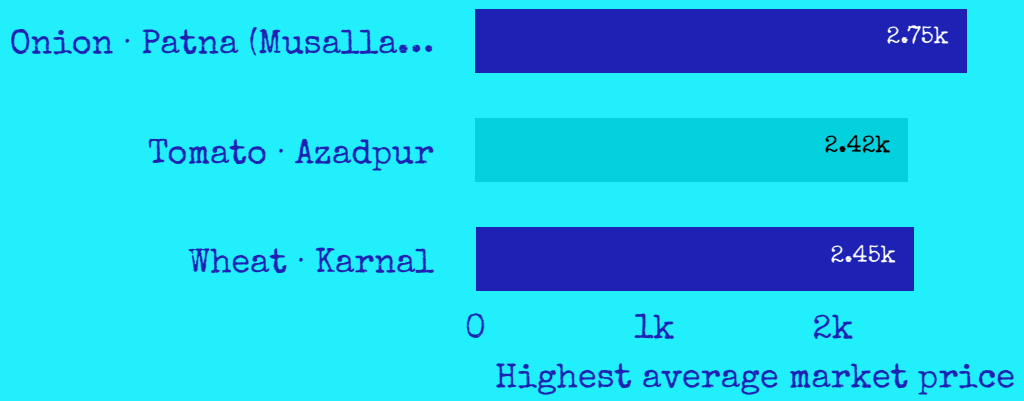
Dashboard evidence

- **Tomato:** CV 45.8%, price spread 7,700, peak-to-floor ratio 14.28
- **Onion:** CV 33.2%, spread 4,320, ratio 5.75
- **Wheat:** CV 9.3%, spread 1,110, ratio 1.57

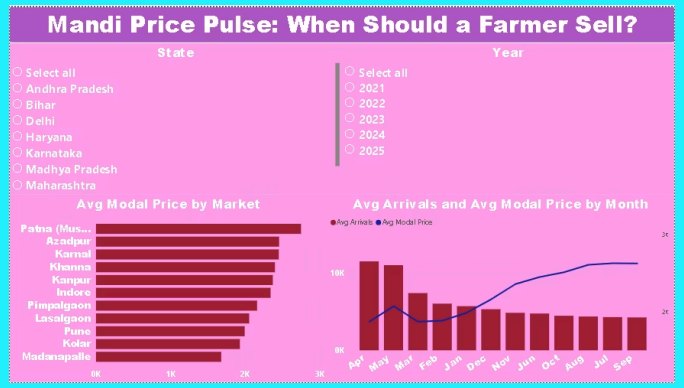
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MARKET CHOICE

MARKET CHOICE CAN MATTER AS MUCH AS MONTH CHOICE



Highest average market by crop, based on the verified analysis.



Most stable markets by crop: Onion—Azadpur CV 29.2%; Tomato—Pune 39.4%; Wheat—Indore 8.5%.

> Supply signal: monthly arrivals and prices show strong negative association for onion -0.996 and tomato -0.998; this does not establish causation.

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